Christian Villanueva

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Functional Test Fixture Engineer completing a B.S. in Computer Engineering in December 2026, with a background spanning production hardware validation, embedded systems, and Linux infrastructure. Focused on integrating electrical hardware, embedded devices and software into functional, testable systems.

EXPERIENCE

QxQ Inc., Fremont, CA *May 2025 – Present*

Functional Test Fixture Engineer (promoted from Test Fixture Technician, May 2026)

- Built, wired, brought up, and validated 50+ functional test fixtures from customer schematics for server, automotive, networking and robotics hardware across 5-48 V power domains.
- Directed development and deployment of a production tool used by three team members, reducing CSV-to-XLS/DRL processing time by approximately 94% while incorporating DFT-based probe sizing.
- Established required pre-shipment validation using continuity checks and customer checklists to catch miswires, loose connectors, missing grounds and incorrect terminations.
- Trained two technicians to independently assemble and verify fixtures, taking on lead responsibility for fixture production and technical escalations.
- Collaborated on-site with client engineers to isolate fixture interconnect faults and resolve communication issues involving SPI and I²C.

TECHNICAL PROJECTS

Server Infrastructure and Application Deployment *May 2025 – Present*

Personal Home Lab

- Selected components and built a TrueNAS SCALE Linux server; manage ZFS datasets and deploy containers using Docker Compose.
- Configured Pi-hole DNS, Nginx Proxy Manager, and Tailscale for reverse proxy routing and remote administration.
- Operated Ubuntu and Debian virtual machines on a Supermicro-based Proxmox host; used IPMI/BMC for remote power control and hardware monitoring.
- Designed and implemented a segmented home network infrastructure using VLANs, DHCP reservations, and firewall rules to isolate management, IoT, and guest traffic.

Project L.U.C.I.A: LiDAR Utility Controlled Imaging Assistant *Jan 2026 – Present*

SJSU Computer Engineering Capstone, Team of 4

- Architected an autonomous mapping platform using Raspberry Pi, ROS 2 and LiDAR, defining hardware integration and web-based monitoring and control.
- Led integration of SLAM and autonomous navigation; validated a functional prototype that explores indoor environments while generating a live map.
- Coordinated integration of NVIDIA Jetson and a ZED 2i stereo camera to add object recognition alongside LiDAR-based mapping.

Project Argus: Multi-Sensor Ethernet Monitor *May 2026 – Present*

- Developed PlatformIO/C++ firmware to initialize and read five DS18B20 temperature probes and an SHT31-D temperature/humidity sensor on an ESP32 using 1-Wire and I²C.
- Integrated W5500 Ethernet hardware and basic initialization; performed staged sensor bring-up and verified the buck converter's 3.3 V output.

EDUCATION

San Jose State University, San Jose, CA *Expected December 2026*

Bachelor of Science in Computer Engineering

Relevant coursework: Circuit Analysis, Electronics for Computing Systems, Power Electronics, RF Circuit Design, Microprocessor Design, Digital Design I & II (Verilog), Operating Systems (C), Embedded Systems (C), CPU Design (Verilog)

Mission College, Santa Clara, CA *Graduated 2024*

A.S. Mathematics | A.A. Liberal Arts, Honors: Magna Cum Laude, Dean's List

TECHNICAL SKILLS

Test & Validation: Electrical test fixture integration/bring-up, DUT interfacing, continuity & signal verification, fault isolation/root-cause analysis, DMM & bench PSU diagnostics, crimping, soldering

Protocols: SPI, I²C, Ethernet, 1-Wire

Programming & Tools: C, C++, Verilog, Git, PlatformIO, Vivado, ROS 2

Systems and Networking: Linux CLI, TrueNAS SCALE, ZFS, Docker Compose, Proxmox, IPMI/BMC, VLANs, DHCP, DNS, firewall configuration

Academic Lab Equipment: Oscilloscopes, function generators